

The Harmonic Mean

A worksheet on average speed, spring systems and parallel resistors

Name: _____ Class: _____ Date: _____

Before attempting this worksheet, explore the interactive page at

https://leeleamacjames.github.io/harmonic_mean_explainer.html

and the PhET spring simulator at

https://phet.colorado.edu/sims/html/hooks-law/latest/hooks-law_en.html

Section A (10) Section B (10) Section C (10) Section D (10) **Total /40**

Section A: The Car Problem — Understanding Average Speed

Scenario. A car drives 60 km from London to Brighton at 60 km/h, then returns the same 60 km at 40 km/h. The interactive page on the website walks through this calculation step by step.

Q1. Before doing any calculation, most people guess the average speed is _____ km/h. Write down why that feels like the right answer in one sentence. [1 mark]

Q2. Show the correct calculation of the car's average speed, following the four steps on the website. Fill in every blank. [3 marks]

Step 1 Time for outward leg: $t_1 = d/v_1 = 60 / \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ hours

Step 2 Time for return leg: $t_2 = d/v_2 = 60 / \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ hours

Step 3 Total distance = _____ km; Total time = _____ hours

Step 4 True average speed = _____ / _____ = _____ km/h

Q3. Explain in one sentence why the arithmetic mean (50 km/h) gives the wrong answer. Your explanation should use the words *time* and *equal*. [1 mark]

Q4. A lorry travels a distance d at speed p , then the same distance at speed q . Write an algebraic expression for the true average speed. (*You may look up the derivation on the website's Section 3.*) [1 mark]

$$\bar{v} = \underline{\hspace{2cm}}$$

Q5. The investigation on the website reveals the true average speed through a sequence of steps. Complete each line for $a = 60$, $b = 40$. [4 marks]

(a) Arithmetic mean: $A = (a + b)/2$ $A = \underline{\hspace{2cm}}$

(b) Geometric mean: $G = \sqrt{ab}$ $G = \underline{\hspace{2cm}}$

(c) $G^2 = \underline{\hspace{2cm}}$ $G^2 = \underline{\hspace{2cm}}$

(d) The ratio G^2/A $= \underline{\hspace{2cm}}$

The website shows that G^2/A equals $\underline{\hspace{2cm}}$ km/h, which is the same as the true average speed found in Q2. This quantity has the name: $\underline{\hspace{2cm}}$.

Section B: The Harmonic Mean — Definition and Calculation

The **harmonic mean** of n numbers is defined as:

$$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}} \quad \text{For two numbers: } H = \frac{2ab}{a+b} = \frac{G^2}{A}$$

where $G = \sqrt{ab}$ is the geometric mean and $A = (a+b)/2$ is the arithmetic mean.

Q6. Calculate the harmonic mean of 60 and 40 using the formula $H = 2ab/(a+b)$, showing full working. [2 marks]

Q7. The website states a chain of inequalities for four different means:

$$Q \geq A \geq G \geq H$$

where Q is the quadratic (root-mean-square) mean. For $a = 60$, $b = 40$, the website gives $Q \approx 50.99$. Arrange the four values 50.99, 50, 48.99, 48 next to the correct mean. [2 marks]

Quadratic (Q)	Arithmetic (A)	Geometric (G)	Harmonic (H)

Q8. Calculate the harmonic mean of the three numbers 10, 20 and 30. Show every step. [2 marks]

Q9. The website notes that for n identical parallel resistors of value R_0 each, $R_{\text{eff}} = R_0/n$. Verify this is consistent with $R_{\text{eff}} = H/n$ by computing H for two identical resistors both equal to R_0 . [2 marks]

Q10. A student claims: “The harmonic mean of two numbers is always less than both of them.” Is this true? Identify the one special case where it is not strictly true and explain. [2 marks]

Section C: PhET Investigation — Hooke's Law Simulator

Open the PhET Hooke's Law simulator:

https://phet.colorado.edu/sims/html/hooks-law/latest/hooks-law_en.html

Use the **Series** and **Parallel** system tabs. Record your readings directly in the spaces below.

Q11. Series system (two springs in a line). Set Top Spring $k_1 = 600$ N/m and Bottom Spring $k_2 = 200$ N/m. Apply a force of 100 N. [4 marks]

- (a) Read the total extension from the simulator: $x_{\text{total}} = \underline{\hspace{2cm}}$ m
- (b) Calculate what you would expect using $x = F/k_{\text{eff}}$, where k_{eff} is found from $1/k_{\text{eff}} = 1/k_1 + 1/k_2$. Show your working.

- (c) Do your calculated and simulated values agree? If not, identify one possible source of discrepancy.

- (d) Complete this sentence by calculating $H(k_1, k_2)$:

"The effective spring constant of a series system equals _____ of the harmonic mean of the individual spring constants. For $k_1 = 600$ and $k_2 = 200$: $H = \underline{\hspace{2cm}}$ N/m, so $k_{\text{eff}} = H/2 = \underline{\hspace{2cm}}$ N/m."

Q12. Parallel system. Now switch to the Parallel tab. Set Left Spring $k_1 = 600$ N/m, Right Spring $k_2 = 200$ N/m, and apply 100 N. [3 marks]

- (a) Read the extension: $x = \underline{\hspace{2cm}}$ m
- (b) The parallel rule is $k_{\text{parallel}} = k_1 + k_2$. Calculate the expected extension and compare with the simulator.

- (c) Is the extension in the parallel system *larger or smaller* than in the series system? Explain in terms of stiffness.

Q13. Exploration question. In the series system, keep $k_2 = 200 \text{ N/m}$ fixed and vary k_1 from 200 N/m to 600 N/m in steps of 100 N/m . Record the total extension for a constant force of 100 N . [3 marks]

$k_1 \text{ (N/m)}$	$k_{\text{eff}} \text{ (N/m)}$	x from sim. (m)	x calculated	As k_1 increases,
200				k_{eff} _____
300				(circle one):
400				increases /
500				decreases /
600				stays constant

Why can the effective series stiffness *never* exceed the softer spring, no matter how stiff k_1 becomes?

Section D: Connecting the Ideas — Springs, Circuits and the Mean

Q14. The table below lists physical systems and the formula for the combined quantity. Identify the type of mean (Arithmetic / Harmonic / neither) that appears in each formula, and state what *physically* adds in each case. [3 marks]

System	Combined formula	Type of mean	What adds physically
Resistors in <i>series</i>	$R = R_1 + R_2$		
Resistors in <i>parallel</i>	$R = \frac{R_1 R_2}{R_1 + R_2}$		
Springs in <i>series</i>	$k = \frac{k_1 k_2}{k_1 + k_2}$		
Springs in <i>parallel</i>	$k = k_1 + k_2$		

Q15. The website states: “The harmonic mean is always pulled towards the smaller value.” For parallel resistors, explain physically why making one resistor very small (nearly a short circuit) pulls R_{eff} close to zero, not close to the arithmetic mean. [3 marks]

Q16. A student sets up two resistors $R_1 = 10\ \Omega$ and $R_2 = 20\ \Omega$ in parallel and measures a current of 1.65 A from a 9 V supply (matching the PhET screenshot shown in class).

- (a) Calculate R_{eff} from Ohm’s law $V = IR$.

- (b) Verify this agrees with the harmonic mean formula $R = R_1 R_2 / (R_1 + R_2)$.

[2 marks]

Q17. The website’s investigation showed that for two numbers a and b :

$$H = \frac{G^2}{A}$$

where $G = \sqrt{ab}$ and $A = (a + b)/2$. Verify this identity algebraically by substituting the expressions for G and A into G^2/A and showing it simplifies to $2ab/(a + b)$. [2 marks]

Q18. Open-ended. The website mentions that the four means obey $Q \geq A \geq G \geq H$, with equality only when $a = b$. In the context of springs in series, what does the equality case ($k_1 = k_2$) mean physically, and what would k_{eff} equal in that case? Verify your answer with a numerical example. [2 marks]

Mark Scheme (for teacher use — do not distribute with student copy)**Section A**

Q1. 50 km/h (arithmetic mean). The instinct is to add and halve.

Q2. Step 1: $t_1 = 60/60 = 1.0$ h. Step 2: $t_2 = 60/40 = 1.5$ h. Step 3: total = 120 km, 2.5 h. Step 4: $\bar{v} = 120/2.5 = 48$ km/h.

Q3. The arithmetic mean assumes equal *time* at each speed, but here equal *distances* are covered, so more time is spent at the slower speed.

Q4. $\bar{v} = 2pq/(p+q)$.

Q5. (a) $A = 50$. (b) $G = \sqrt{2400} \approx 48.99$. (c) $G^2 = 2400$. (d) $G^2/A = 2400/50 = 48$. Named: **Harmonic mean**.

Section B

Q6. $H = 2 \times 60 \times 40/(60 + 40) = 4800/100 = 48$.

Q7. $Q \approx 50.99$; $A = 50$; $G \approx 48.99$; $H = 48$.

Q8. $H = 3/(1/10 + 1/20 + 1/30) = 3/(6/60 + 3/60 + 2/60) = 3/(11/60) = 180/11 \approx 16.36$.

Q9. $H(R_0, R_0) = 2R_0^2/(R_0 + R_0) = 2R_0^2/(2R_0) = R_0$. Then $R_{\text{eff}} = H/2 = R_0/2$, which matches R_0/n for $n = 2$. ✓

Q10. False: equality $H = a = b$ when $a = b$.

Section C

Q11. (a) $x = 0.667$ m (PhET). (b) $1/k_{\text{eff}} = 1/600 + 1/200 = 4/600$; $k_{\text{eff}} = 150$ N/m; $x = 100/150 = 0.667$ m. ✓ (c) Should agree to 3 s.f. (d) Half; $H = 2 \times 600 \times 200/800 = 300$ N/m; $k_{\text{eff}} = 150$ N/m.

Q12. (a) $x = 0.125$ m (PhET). (b) $k_{\text{par}} = 800$ N/m; $x = 100/800 = 0.125$ m. ✓ (c) Smaller extension — parallel system is stiffer ($k = 800 > 150$).

Q13. Table: $k_{\text{eff}} = 100, 120, 133, 143, 150$ N/m (rising toward 200); $x = 1.00, 0.833, 0.75, 0.70, 0.667$ m. k_{eff} increases but asymptotes to 200 (the fixed soft spring): the harmonic mean formula means the softer spring always limits the series stiffness. No matter how rigid k_1 , the $1/k_1$ term vanishes but $1/k_2 = 1/200$ remains, so $k_{\text{eff}} \rightarrow k_2$.

Section D

Q14. Series resistors: Arithmetic (resistances add); Parallel resistors: Harmonic/2 (conductances $1/R$ add); Series springs: Harmonic/2 (compliances $1/k$ add); Parallel springs: Arithmetic (stiffnesses add).

Q15. A very small $R_1 \approx 0$ means conductance $1/R_1 \rightarrow \infty$, so $1/R_{\text{eff}} = 1/R_1 + 1/R_2 \rightarrow \infty$, giving $R_{\text{eff}} \rightarrow 0$. A direct short-circuit path carries virtually all the current, bypassing R_2 entirely.

Q16. (a) $R = V/I = 9/1.65 \approx 5.45 \Omega$. (b) $10 \times 20/(10 + 20) = 200/30 = 6.67 \Omega$. Wait — re-check: $200/30 \neq 5.45$. Correct: the three-resistor parallel (10, 20, 30) gives $60/11 \approx 5.45 \Omega$, matching the PhET image. For two resistors $R_1 R_2/(R_1 + R_2) = 200/30 \approx 6.67 \Omega$.

Q17. $G^2/A = ab / \frac{a+b}{2} = \frac{ab \cdot 2}{a+b} = \frac{2ab}{a+b} = H$. ✓

Q18. When $k_1 = k_2 = k_0$ both springs are identical. $k_{\text{eff}} = k_0^2/(2k_0) = k_0/2$ — exactly half the spring constant of either. Example: $k_0 = 200$ N/m gives $k_{\text{eff}} = 100$ N/m. This is the equality case of the harmonic-mean inequality: $H(k_0, k_0) = k_0$.